

ELAINE HUANG

huang12211@gmail.com | (438) 886-9528 | [LinkedIn](#) | [Portfolio](#) | [GitHub](#) | Toronto, ON

PROFILE

Customer-facing product engineer with 7 years defining, designing and delivering high-risk software end-to-end in regulated medical devices, including web development and proof of concept prototypes that secured \$2M in funding. Ships consumer web apps solo, design through deployment, in TypeScript/React and Python. Looking for Full-stack positions so that I can build full-time.

PROJECTS

Elaine's Easicipes | [GitHub](#) | Live at elaineseasicipes.com | Ongoing
0-to-1 AI-native full-stack web application designed and built end-to-end from mockups to implementation and deployment. Includes "Pitaya Pal", an in-app AI assistant that makes recommendations using RAG over the recipe database. (*Figma, Next.js, React, TypeScript, Tailwind CSS, SQLite, Vercel AI SDK, Langfuse, Railway, API Design*)

Automatic Test Case Generation - Multi-Agent System | [GitHub](#) | Jun 2026
Designed a RAG + ReAct agent pipeline that retrieves context from proprietary specifications using Chroma (vector DB) and generates structured test cases with human-in-the-loop feedback. (*Python, LangGraph, Ollama, Chroma, RAG, ReAct, HITL*)

Literature Review Gemini AI Assistant | [GitHub](#) | Sep 2025
An LLM-powered retrieval pipeline using LangChain and Gemini to perform automated literature reviews on medical device risk documentation; extracted data from research documents with 90% accuracy on known benchmark cases. (*Python, LangChain, Gemini, Document Retrieval, LLM Orchestration, Prompt Engineering*)

WORK EXPERIENCE

Zimmer Biomet, *Montreal QC* Sept 2019 – present

Senior Systems Designer / Product Owner

- Translated surgeon user research into mockups and engineering specifications for 8 features across 4 product releases, aligning clinical, engineering, marketing and regulatory stakeholders under feasibility, risk, and schedule constraints. (*Figma*)
- Standardized the interface across 7 product lines by co-developing a Figma design system (reusable component library, system design patterns, and GUI standards) which was adopted by 14 designers, supporting consistent use-safety behavior across the portfolio. (*Figma*)
- Built AI tools for literature reviews, documentation and GUI testing, cutting execution time by 30% over 6 months. (*LangChain, LangGraph, GitHub Copilot, Python*)
- Drove organization-wide adoption of these low-code AI workflows through enablement sessions, 1:1 hands-on support, and a reusable library of prompts, skills, and workflow patterns now used by 8 teams.
- Built automated Python test suites for features I specified end-to-end, cutting regression test execution time from 2 days to 20 hours per release. (*Python*)

Innovation Program Coordinator

- Identified market gaps through market research and pitched 4 product proposals; 3 secured a place in the product roadmap.
- Prototyped an augmented reality surgery device that earned a granted US patent, shipping a working MVP in 6 weeks to secure \$2M in board-approved funding. (*SolidWorks, C#*)
- Built partnerships with local universities, hospitals, and imaging centers, securing access to imaging and clinical testing facilities that saved \$500K annually in study costs.

SKILLS

Front-end: TypeScript, React, Next.js, HTML/CSS, Tailwind CSS, Responsive UI

UX, UI: Figma, Lovable, Miro, User research, A/B testing

AI / ML: Agentic workflows, LangChain, LangGraph, RAG, Embeddings, Vector databases (Chroma), Prompt engineering, human-in-the-loop (HITL), LLM orchestration, Ollama, Gemini, Claude Code, GitHub Copilot

Backend / Data: Python, SQL, REST APIs, JWT authentication, Matlab, C#

Deployment: Railway, Git/GitHub, CI/CD pipelines, Secrets management, Persistent volumes, Database migrations

Testing & Automation: Squish, Cucumber, Selenium

Product: Lean-Agile, JIRA, Confluence

Languages: Native English & French; Basic Chinese

EDUCATION & CERTIFICATIONS

Applied Data Science Program: Leveraging AI for Effective Decision-Making

MIT Professional Education

Bachelor of Applied Science, in Biomedical Engineering, Co-op

University of Waterloo, Waterloo, ON

Graduated with Distinction (GPA: 3.87/4.00) | Baylis Medical Award for Capstone Design Project (\$5k)